

Alejandro Dopico-Castro

PhD Researcher in Frugal AI | Continuous & Federated Learning

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Education

PhD	University of A Coruña (UDC) , Computer Science Funded PhD on Frugal and Sustainable AI (continual, federated, few-shot learning).	Mar 2025 – present
MSc	University of A Coruña (MIA-UDC) , Artificial Intelligence • Thesis: Efficient Single-Step Framework for Incremental Class Learning	Sept 2023 – Feb 2025
BSc	University of A Coruña , Computer Science • Thesis: Object detection for sports images	Sept 2019 – June 2023

Experience

LIDIA Group, University of A Coruña , PhD Researcher • Research on frugal AI methods (continual, federated, few-shot learning) for resource-constrained systems. • Development of adaptive learning modules within the EU PILLAR-Robots project.	A Coruña, Spain Mar 2025 – present 1 year 7 months
INESC TEC, University of Porto , Research Stay (Visiting Researcher) • Research stay with João Gama (University of Porto) and Pedro Henriques Abreu (University of Coimbra). • Collaboration on deep learning and continual learning for resource-efficient settings.	Porto, Portugal Mar 2026 – May 2026 3 months
LIDIA Group, University of A Coruña , Research Support Technician (MSc Internship) • Developed class-incremental learning models for dynamic environments within the PILLAR-Robots project. • Work evolved into current PhD research on Frugal AI.	A Coruña, Spain Jan 2024 – Feb 2025 1 year 2 months
Cinfo Company , AI Software Developer • Designed and optimized deep learning models (YOLO, SSD) for automated sports production. • Integrated C++ real-time inference with Python-based training pipelines.	A Coruña, Spain Oct 2022 – June 2023 9 months

Publications

FedHENet: A Frugal Federated Learning Framework for Heterogeneous Environments Single-round, hyperparameter-free federated image classification with homomorphic encryption. Alejandro Dopico-Castro, Oscar Fontenla-Romero, Bertha Guijarro-Berdiñas, Amparo Alonso-Betanzos, Iván Pérez Digón arxiv.org/abs/2602.13024 (ESANN)	2026
Closing the Alignment-Maturity Gap in Federated Prototype Learning Scheduled alignment and geometry-driven separation for federated prototype learning under heterogeneity. Mario Casado-Diez, Alejandro Dopico-Castro, Verónica Bolón-Canedo, Bertha Guijarro-Berdiñas arxiv.org/abs/2606.02172 (arXiv)	2026

CIFNet: An Analytic Neural Learning Framework for Efficient and Calibrated Class-Incremental Learning

2025

Closed-form class-incremental learning with calibrated analytic updates and lower energy cost.

Alejandro Dopico-Castro, Oscar Fontenla-Romero, Bertha Guijarro-Berdiñas, Amparo Alonso-Betanzos

[10.48550/arXiv.2509.11285](https://arxiv.org/abs/10.48550/arXiv.2509.11285) (arXiv)

Skills

ML Frameworks: PyTorch, TensorFlow, Scikit-Learn, OpenCV, HuggingFace

Programming: Python, SQL, C++, Shell

Tools: Docker, Git, MLFlow, DVC, CI/CD

Cloud: AWS (EC2, S3)

Research Areas: Continual Learning, Federated Learning, Frugal AI, Edge AI, Computer Vision